

MINGJIA SHI

PH.D. STUDENT W/ M.E.

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INTERNSHIPS

AI Research & Development Intern at Nokia
Sunnyvale, California, U.S.

May–Aug. 2026

- Research focus: long-context compression and robustness. On-site internship.

Research Intern in Houmoai | Nanjing, China

Jan. 2025 - July 2025

- *Project.* Develop analyses methods for resource preserving.
- Prepare reports and presentations about research findings.
- Research: resource-preserving analysis methods and parameter modeling.

Research Assistant Intern in HPC-AI Lab, NUS | Singapore

June 2023 - Jan. 2025

- *Project.* Develop analyses methods for resource preserving.
- Write paper and prepare presentations about research findings.
- Research: resource-preserving analysis methods.

Research Intern in Lenovo (Beijing) | Beijing, China

Nov. 2023 - Jan. 2024

- *Project.* Develop analyses methods for decentralized data.
- Prepare reports and presentations about research findings.

Research Intern in Lenovo (Chengdu) | Chengdu, China

June 2018 - Aug. 2018

- *Project.* Deployment of decentralized data analyses system.
- Test usage and report research findings about the decentralized system.

EDUCATION

School of Engineering and Applied Science, Uni. of Virginia Charlottesville, VA, U.S.

Ph.D. in Computer Engineering

Aug. 2025 - 2030 (anticipated)

- Advisor: Professor Jundong Li
- Research: learning theory, efficient AI, and multimodal learning.
- Coursework: mathematical foundations of machine learning.

College of Computer Science, Sichuan University

Chengdu, China

Master of Engineering in Artificial Intelligence

Sep. 2021 - June 2024

- Advisor: professor Jiancheng Lv
- Research: resource preserving decentralized data analyses.
- Course category: research analysis, math. and data science.

College of Computer Science, Sichuan University

Chengdu, China

Bachelor of Engineering in Computer Science

Sep. 2015 - June 2019

- Advisor: professor Jiancheng Lv
- Research: decentralized optimization and data science.
- Course category: foundational knowledge about computer, programming, math. and data analysis methods.

HONORS

- **Honorable Undergraduate Award**, Bachelors Degree in Sichuan University June 2019

SKILLS

Languages: Chinese, English.

Programming: Python. **Research:** learning theory, efficient AI, multimodal learning, and data analysis.

ACADEMIC SERVICES

Conference reviewer: NeurIPS, ICLR, ICML, CVPR, ICCV, ACM MM, ICPP, ICASSP, and ICONIP.

Journal reviewer: IEEE TNNLS, TCSVT, TPDS, TKDE, Neurocomputing, and others.

First-Author / Co-First-Author Papers

1. **Mingjia Shi**, Yinhan He, Yaochen Zhu, Jundong Li. Saliency-Aware Multi-Route Thinking: Grounding and Reasoning on Vision-Language Agents. *NeurIPS*, 2026.
Brief: training-free principle selection and parallel reasoning for visually grounded vision-language agents.
2. **Mingjia Shi**, Shuo Wang, Xiaobo Wang, Sifan Zhou, Kai Wang, Tianyu Fu, Chenxu Zhao, Anyang Su, Ping Jiang, Minghui Wu. Dive into the Implicit Biases of Low-rank Vision-language Alignment. *ECCV*, 2026.
Brief: analyzing why low-rank vision-language alignment reduces computation and preserves visual representation structure.
3. Kai Wang*, **Mingjia Shi***, Yukun Zhou, Zhihang Yuan, Yuzhang Shang, Xiaojiang Peng, Hanwang Zhang, Yang You. A Closer Look at Time Steps is Worthy of Triple Speed-Up for Diffusion Model Training. *CVPR*, 2025.
Brief: accelerating building popular generative models.
4. **Mingjia Shi***, Ruihan Lin*, Xuxi Chen, Yuhao Zhou, Zezhen Ding, Pingzhi Li, Tong Wang, Kai Wang, Zhangyang Wang, Jiheng Zhang, Tianlong Chen. Make Optimization Once and for All with Fine-grained Guidance. *arXiv*, 2025.
Brief: modeling parameter space during optimization.
5. **Mingjia Shi**, Yuhao Zhou, Ruiji Yu, Zekai Li, Zhiyuan Liang, Xuanlei Zhao, Xiaojiang Peng, Shanmukha Ramakrishna Vedantam, Wangbo Zhao, Kai Wang, Yang You. Faster Vision Mamba is Rebuilt in Minutes via Merged Token Re-training. *arXiv*, 2024.
Brief: accelerating popular sequence model.
6. **Mingjia Shi**, Yuhao Zhou, Kai Wang, Huaizheng Zhang, Shudong Huang, Qing Ye, Jiancheng Lv. PRIOR: Personalized Prior for Reactivating the Information Overlooked in Federated Learning. *NeurIPS*, 2023.
Brief: information overlooked issue is proposed in decentralized data analysis.
7. **Mingjia Shi**, Yuhao Zhou, Qing Ye, Jiancheng Lv. Unconstrained Feature Model and Its General Geometric Patterns in Federated Learning: Local Subspace Minority Collapse. *ICONIP*, 2023.
Brief: local subspace minority collapse in decentralized data analysis

Co-Authored Papers

8. Zekai Li, Xinhao Zhong, Samir Khaki, Zhiyuan Liang, Yuhao Zhou, **Mingjia Shi**, Ziqiao Wang, Xuanlei Zhao, Wangbo Zhao, Ziheng Qin, Mengxuan Wu, Pengfei Zhou, Haonan Wang, David Junhao Zhang, Jia-Wei Liu, Shaobo Wang, Dai Liu, Linfeng Zhang, Guang Li, Kun Wang, Zheng Zhu, Zhiheng Ma, Joey Tianyi Zhou, Jiancheng Lv, Yaochu Jin, Peihao Wang, Kaipeng Zhang, Lingjuan Lyu, Yiran Huang, Zeynep Akata, Zhiwei Deng, Xindi Wu, George Cazenavette, Yuzhang Shang, Justin Cui, Jindong Gu, Qian Zheng, Hao Ye, Shuo Wang, Xiaobo Wang, Yan Yan, Angela Yao, Mike Zheng Shou, Tianlong Chen, Hakan Bilen, Baharan Mirzasoleiman, Manolis Kellis, Konstantinos N Plataniotis, Zhangyang Wang, Bo Zhao, Yang You, Kai Wang. DD-Ranking: Rethinking the Evaluation of Dataset Distillation. *NeurIPS*, 2026. **Oral presentation**.
Brief: benchmark of data distillation.
9. Zhenyu Lei, **Mingjia Shi**, Xingbo Fu, Haoyu He, Qi R. Wang, Jundong Li. Grounding Memory Summarization in Utility Intent. *EMNLP*, 2026.
Brief: utility-aware memory summarization through self-distillation and retriever alignment.
10. Yinhan He, Yaochen Zhu, **Mingjia Shi**, Wendy Zheng, Lin Su, Xiaoqing Wang, Qi Guo, Jundong Li. IAPO: Information-Aware Policy Optimization for Token-Efficient Reasoning. *ICML*, 2026.
Brief: information-aware reinforcement learning for accurate and token-efficient reasoning.
11. Mengxuan Wu, Xinhao Zhong, Penghao Wang, Ziheng Qin, Jimeng Zhang, Wangbo Zhao, **Mingjia Shi**, Zekai Li, Hao Ye, Panpan Zhang, Shuo Wang, Yang You, Kai Wang, Dawei Yang. QuantGen: Parameter Generation for Controllable Model Quantization. *ICLR 2026 submission*, 2025.
Brief: controllable generation of quantized model parameters for different deployment requirements.
12. Tianyu Fu, Anyang Su, Chenxu Zhao, Hanning Wang, Minghui Wu, Zhe Yu, Fei Hu, **Mingjia Shi**, Wei Dong, Jiayao Wang, Yuyang Chen, Ruiyang Yu, Siran Peng, Menglin Li, Nan Huang, Haitian Wei, Jiawei Yu, Yi Xin, Xilin Zhao, Kai Gu, Ping Jiang, Sifan Zhou, Shuo Wang. Mano Technical Report. *arXiv*, 2025.
Brief: LLM-based GUI agent.

13. Zhiyuan Liang, Dongwen Tang, Yuhao Zhou, Xuanlei Zhao, **Mingjia Shi**, Wangbo Zhao, Zekai Li, Peihao Wang, Konstantin Schürholt, Damian Borth, Michael M. Bronstein, Yang You, Zhangyang Wang, Kai Wang. Drag-and-Drop LLMs: Zero-Shot Prompt-to-Weights. *NeurIPS*, 2025.
Brief: generate models’ weights by instructions.
14. Jindi Lv, Yuhao Zhou, **Mingjia Shi**, Zhiyuan Liang, Panpan Zhang, Xiaojiang Peng, Wangbo Zhao, Zheng Zhu, Jiancheng Lv, Qing Ye, Kai Wang. Pruning-Robust Mamba with Asymmetric Multi-Scale Scanning Paths. *NeurIPS*, 2025.
Brief: Pruning tokens in Mamba for acceleration with robustness.
15. Ziqiao Wang, Wangbo Zhao, Yuhao Zhou, Zekai Li, Zhiyuan Liang, **Mingjia Shi**, Xuanlei Zhao, Pengfei Zhou, Kaipeng Zhang, Zhangyang Wang, Kai Wang, Yang You. REPA Works Until It Doesn’t: Early-Stopped, Holistic Alignment Supercharges Diffusion Training. *NeurIPS*, 2025.
Brief: no need to training diffusion models with REPA through all the process.
16. Xinghao Wu, Jianwei Niu, Xuefeng Liu, **Mingjia Shi**, Guogang Zhu, Shaojie Tang. Tackling Feature-Classifer Mismatch in Federated Learning via Prompt-Driven Feature Transformation. *NeurIPS*, 2025.
Brief: feature mismatching issue in decentralized data analysis.
17. Sifan Zhou*, Shuo Wang*, Zhihang Yuan*, **Mingjia Shi**, Yuzhang Shang, Dawei Yang. GSQ-Tuning: Group-Shared Exponents Integer in Fully Quantized Training for LLMs On-Device Fine-tuning. *ACL Findings*, 2025.
Brief: resource-preserving application of popular models.
18. Yuhao Zhou, Yuxin Tian, Jindi Lv, **Mingjia Shi**, Yuanxi Li, Qing Ye, Shuhao Zhang, Jiancheng Lv. Ferret: An Efficient Online Continual Learning Framework under Varying Memory Constraints. *CVPR*, 2025.
Brief: memory efficient methods for streaming applications.
19. Yuhao Zhou*, Yuxin Tian*, **Mingjia Shi**, Yuanxi Li, Yanan Sun, Qing Ye, Jiancheng Lv. E-3SFC: Communication-Efficient Federated Learning with Double-way Features Synthesizing. *TNNLS*, 2025.
Brief: communication-efficient algorithm for decentralized data analysis.
20. Haixin Gao, **Mingjia Shi**, Yuhao Zhou, Deng Xiong, Qing Ye, Jiancheng Lv. FedSH: Tackling Staleness by Scheduling High-order Approximation in Asynchronous Federated Learning. *IJCNN*, 2025.
Brief: scheduled high-order approximation for reducing gradient-staleness errors.
21. Yuhao Zhou, **Mingjia Shi**, Yuxin Tian, Qing Ye, Jiancheng Lv. DeFTA: A Plug-and-Play Peer-to-Peer Decentralized Federated Learning Framework. *InfoSci*, 2024.
Brief: balancing algorithm on decentralized data analysis.
22. Yuhao Zhou, **Mingjia Shi**, Yuxin Tian, Yuanxi Li, Qing Ye, Jiancheng Lv. Federated CINN Clustering for Accurate Clustered Federated Learning. *ICASSP*, 2024.
Brief: communication-efficient decentralized data analysis algorithm with clustering.
23. Yuhao Zhou, **Mingjia Shi**, Yuanxi Li, Yanan Sun, Qing Ye, Jiancheng Lv. Communication-efficient Federated Learning with Single-Step Synthetic Features Compressor for Faster Convergence. *ICCV*, 2023.
Brief: communication-efficient analysis algorithm of decentralized data distillation.
24. Qing Ye, Yuhao Zhou, **Mingjia Shi**, Yanan Sun, Lv Jiancheng. DLB: A Dynamic Load Balance Strategy for Distributed Training of Deep Neural Networks. *TETCI*, 2022.
Brief: balancing algorithm motivated by different batch size of data.
25. Qing Ye, Yuhao Zhou, **Mingjia Shi**, Jiancheng Lv. FLSGD: Free Local SGD with Parallel Synchronization. *JoSc*, 2022.
Brief: algorithm accelerating distributed data analysis.